Per-criterion metrics: BettaTool vs MFLD-Net

Test split, 247 held-out images (1,230 criterion assessments; caudal-body ratio has 242). Fault is the positive class. Ground truth is the IBC rule applied to the human-labelled keypoints.

Forced: the model answers every case. MFLD-Net has no gate, so it is always forced. BettaTool (forced) is the like-for-like comparison. **Gate**: BettaTool defers to a judge when its predicted keypoint error is not below the TSI, so its metrics cover only the cases it decides (the Decided column). **Not calibrated** uses the TSI rule as written (scale = 1.0). **Calibrated** (the deployed gate) multiplies the predicted error by a per-criterion scale fitted on the validation split, which makes it defer more. n/a means undefined (no predicted or no actual Faults). Each confusion matrix shows actual label by row and predicted label by column.

Source: backend/analysis/make_confusion_matrix_mfld.py, outputs in backend/confusion_matrices_out/.

Summary

System	Decided	Accuracy	Precision	Recall	F1
MFLD-Net (forced)	1230 (100.0%)	0.937	0.891	0.859	0.875
BettaTool (forced)	1230 (100.0%)	0.957	0.925	0.904	0.914
BettaTool (gate, not calibrated)	1049 (85.3%)	0.992	0.995	0.969	0.982
BettaTool (gate, calibrated)	**887 (72.1%)**	**0.997**	**1.000**	**0.979**	**0.990**

Caudal spread angle

Cases scored: 247. Actual Faults: 247. Actual Pass: 0.

System	Decided	Accuracy	Precision	Recall	F1
MFLD-Net (forced)	247 (100.0%)	1.000	1.000	1.000	1.000
BettaTool (forced)	247 (100.0%)	1.000	1.000	1.000	1.000
BettaTool (gate, not calibrated)	203 (82.2%)	1.000	1.000	1.000	1.000
BettaTool (gate, calibrated)	**133 (53.8%)**	**1.000**	**1.000**	**1.000**	**1.000**

System	Actual	Pred Fault	Pred Pass	Defer
MFLD-Net (forced)	Fault	247	0	–
	Pass	0	0	–
BettaTool (forced)	Fault	247	0	–
	Pass	0	0	–
BettaTool (gate, not calibrated)	Fault	203	0	44
	Pass	0	0	0
BettaTool (gate, calibrated)	Fault	133	0	114
	Pass	0	0	0

Note: Every test fish is a Fault under the IBC bands, so there are no Pass cases. Accuracy, precision and recall are trivially 1.000 for any system that answers Fault, and this criterion does not separate the models.

Dorsal fin / body ratio

Cases scored: 247. Actual Faults: 43. Actual Pass: 204.

System	Decided	Accuracy	Precision	Recall	F1
MFLD-Net (forced)	247 (100.0%)	0.834	0.531	0.395	0.453
BettaTool (forced)	247 (100.0%)	0.879	0.710	0.512	0.595
BettaTool (gate, not calibrated)	163 (66.0%)	0.982	1.000	0.750	0.857
BettaTool (gate, calibrated)	**121 (49.0%)**	**0.992**	**1.000**	**0.833**	**0.909**

System	Actual	Pred Fault	Pred Pass	Defer
MFLD-Net (forced)	Fault	17	26	–
	Pass	15	189	–
BettaTool (forced)	Fault	22	21	–
	Pass	9	195	–
BettaTool (gate, not calibrated)	Fault	9	3	31
	Pass	0	151	53
BettaTool (gate, calibrated)	Fault	5	1	37
	Pass	0	115	89

Anal fin / body ratio

Cases scored: 247. Actual Faults: 11. Actual Pass: 236.

System	Decided	Accuracy	Precision	Recall	F1
MFLD-Net (forced)	247 (100.0%)	0.923	0.214	0.273	0.240
BettaTool (forced)	247 (100.0%)	0.960	0.533	0.727	0.615
BettaTool (gate, not calibrated)	219 (88.7%)	0.995	1.000	0.750	0.857
BettaTool (gate, calibrated)	**196 (79.4%)**	**0.995**	**1.000**	**0.500**	**0.667**

System	Actual	Pred Fault	Pred Pass	Defer
MFLD-Net (forced)	Fault	3	8	–
	Pass	11	225	–
BettaTool (forced)	Fault	8	3	–
	Pass	7	229	–
BettaTool (gate, not calibrated)	Fault	3	1	7
	Pass	0	215	21
BettaTool (gate, calibrated)	Fault	1	1	9
	Pass	0	194	42

Note: Only 11 actual Faults: low support.

Caudal fin / body ratio

Cases scored: 242. Actual Faults: 3. Actual Pass: 239.

System	Decided	Accuracy	Precision	Recall	F1
MFLD-Net (forced)	242 (100.0%)	0.983	0.400	0.667	0.500
BettaTool (forced)	242 (100.0%)	0.996	0.750	1.000	0.857
BettaTool (gate, not calibrated)	240 (99.2%)	1.000	1.000	1.000	1.000
BettaTool (gate, calibrated)	**236 (97.5%)**	**1.000**	**1.000**	**1.000**	**1.000**

System	Actual	Pred Fault	Pred Pass	Defer
MFLD-Net (forced)	Fault	2	1	–
	Pass	3	236	–
BettaTool (forced)	Fault	3	0	–
	Pass	1	238	–
BettaTool (gate, not calibrated)	Fault	3	0	0
	Pass	0	237	2
BettaTool (gate, calibrated)	Fault	3	0	0
	Pass	0	233	6

Note: Only 3 actual Faults: precision, recall and F1 change a lot with a single case. Low support.

Anal fin length / width ratio

Cases scored: 247. Actual Faults: 9. Actual Pass: 238.

System	Decided	Accuracy	Precision	Recall	F1
MFLD-Net (forced)	247 (100.0%)	0.947	0.000	0.000	n/a
BettaTool (forced)	247 (100.0%)	0.951	0.333	0.333	0.333
BettaTool (gate, not calibrated)	224 (90.7%)	0.982	0.000	0.000	n/a
BettaTool (gate, calibrated)	**201 (81.4%)**	**0.995**	**n/a**	**0.000**	**n/a**

System	Actual	Pred Fault	Pred Pass	Defer
MFLD-Net (forced)	Fault	0	9	–
	Pass	4	234	–
BettaTool (forced)	Fault	3	6	–
	Pass	6	232	–
BettaTool (gate, not calibrated)	Fault	0	3	6
	Pass	1	220	17
BettaTool (gate, calibrated)	Fault	0	1	8
	Pass	0	200	38

Note: Only 9 actual Faults: low support. For the deployed gate, precision and F1 are n/a because it predicted no Faults (it deferred 8 of the 9 and called 1 Pass).

Overall (pooled): confusion matrix

System	Actual	Pred Fault	Pred Pass	Defer
MFLD-Net (forced)	Fault	269	44	–
	Pass	33	884	–
BettaTool (forced)	Fault	283	30	–
	Pass	23	894	–
BettaTool (gate, not calibrated)	Fault	218	7	88
	Pass	1	823	93
BettaTool (gate, calibrated)	Fault	142	3	168
	Pass	0	742	175

Pooled over all 1,230 assessments. The calibrated gate decides 887 (72.1%) and makes 3 wrong decisions, all actual Faults called Pass; it never calls an actual Pass a Fault. The not-calibrated gate decides 1,049 (85.3%) and makes 8 wrong decisions (7 Faults missed, 1 good fish failed).